Malek Swilam

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Education

George Mason University, Honors College — Fairfax, VA
B.S. Computer Engineering — *Aug. 2026 – May 2030 (Expected)*

- **Relevant Coursework (GMU and other institutions):** Object-Oriented Programming, Data Structures & Algorithms, Introduction to Electrical & Computer Engineering, Calculus 3, Physics 1 & 2, Introduction to Engineering

Experience

Machine Learning Intern — Jun. – Aug. 2026
Yanez.ai — *Remote*

- Built a computer vision ensemble for face anti-spoofing, combining a gradient-boosted model on **50+** engineered image cues with a CNN-based depth model, cutting error **72%** versus the deployed rule-based system on unseen data
- Designed a rank-based score fusion for models with incomparable output scales, catching **100%** of real attacks that had bypassed the production system
- Boosted throughput **500%+** on an adversarial document tampering pipeline with multi-threading and a load-aware scheduler, and cut cost per image **85–93%** by replacing general-purpose generative models with a dedicated face-swap model

Engineering Intern — May 2026
Alarm.com — *Tysons Corner, VA*

- Co-built a Flask lab portal for the hardware QA team with QR-labeled inventory, equipment checkout, and admin roles, replacing manual tracking of lab equipment
- Soldered buck converters and embedded boards, provisioned Raspberry Pi units, and integrated locks, sensors, cameras, and thermostats into automated hardware testing

Machine Learning Research Intern — Jun. – Sep. 2025
George Mason University, ASSIP — *Fairfax, VA (Hybrid)*

- Built an ML pipeline for short-term rainfall forecasting, training and comparing **LSTM**, **XGBoost**, and **ensemble** models on MAE and RMSE; published in *GMU Journal of Student-Scientists' Research*
- Ensembled them to outperform both on every metric, landing **96%+** of hourly predictions within ±1mm of actual

Programming Lead — 2023 – 2026
FTC Robotics, Team 369 (Oakton) — *Vienna, VA*

- Led all robot software development in Java across two varsity seasons; built an autonomous localization pipeline fusing wheel odometry with IMU, with AprilTag detection for field-relative pose correction and motion profiling, ranking in the **top 4% of 7,000+ teams internationally** for autonomous scoring
- Designed multi-subsystem PIDF control (turret, flywheel, pose stabilization) and a Limelight-based vision targeting system; reached **State Championship** playoffs both seasons and **captained a playoff alliance** at the international **World Championship**

Projects

OCRadar: Oral Cancer Detection App | *Python, Azure AI Vision, Flutter, Dart* — ocradar.com

- As **COO and lead Android/Windows developer**, co-built a cross-platform app that screens for early signs of oral cancer using a CNN image classifier on Azure AI Custom Vision with **98% accuracy**; featured by **WTOP News**, **Korea Times**, and Vision Times China

Voice-Controlled Robotics Agent | *Python, MuJoCo, Anthropic API, FastAPI, Whisper* — malekswilam.dev/robot-agent

- Built a voice-controlled manipulation agent for a Franka Panda arm in MuJoCo where an LLM decomposes natural-language commands into sub-goals via chain-of-thought reasoning, detects mid-task failures, and replans autonomously
- Solved robot motion via inverse kinematics and streamed the live simulation through a FastAPI WebSocket portal; cut LLM input cost per call ∼**99%** with prompt caching

Desk Calendar Agent | *Python, Raspberry Pi 5, Local LLM, Google Calendar API* — malekswilam.dev/calendar-agent

- Built an agent that turns a texted or spoken instruction into a Google Calendar event using a 4-bit quantized 3B-parameter model (Qwen2.5) running **fully on-device** on a Raspberry Pi 5, with deterministic checks that blocked **100%** of destructive actions the model proposed in testing
- Cut on-device prompt processing time **94%** (∼100s to ∼6s) with prefix caching; wired the e-ink display to the Raspberry Pi 5, wrote the rendering and display-refresh layer, and selected the hardware components

Technical Skills

Languages: Java, Python, Dart, JavaScript, HTML/CSS
Technologies: PyTorch, XGBoost, NumPy, FastAPI, Flask, Flutter, MuJoCo, Azure AI Vision, Anthropic API, Ollama
Hardware: Soldering, Embedded Linux, Raspberry Pi, Sensor Fusion & PIDF Control
Tools: Git/GitHub, Android Studio, Claude Code, Codex
Honors: Eagle Scout; FTC World Championship Alliance Captain; T.R. Cook Award of Excellence in Technology (2025)
Concepts: Machine Learning, Computer Vision, Deep Learning, Data Structures & Algorithms, Ensemble Learning, Agentic AI, Model Evaluation, Edge Inference